Physics-Informed Reconstruction of Unmeasurable Cardiac Fields from Real Data: Electrocardiographic Imaging and 4D-Flow Aortic Pressure, with Honest Nulls

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Abstract*—Two clinically decisive cardiac quantities cannot be measured where they are needed: the heart-surface potential map (measured only with an invasive cage) and the intravascular pressure field (measured only with a catheter). Both are tied by a governing equation to a quantity that *can* be measured non-invasively, the body-surface potential and the 4D-flow velocity, so both can be reconstructed. We report a real-data-only study of this reconstruction across two distinct physics domains, with a strict discipline: every case fits a real measured signal and is validated against a real gold standard or an exact analytic gate; there is no synthetic ground truth, because a network that only re-solves an equation a classical solver already solves answers no clinical question. *Case 1 (electrocardiographic imaging, quasi-static volume conduction).* On real EDGAR experiments reconstructed by an identical pipeline with no per-heart retuning, a forward operator on the real geometry with a graph-Laplacian prior and a deep-ensemble node uncertainty recovers heart-surface potentials at a relative error of 0.54–0.65 and spatial correlation 0.72–0.85 against the measured cage, with $\approx 90\%$ of nodes inside two standard deviations, and paced beats reconstructing better than sinus as physically expected. A full boundary-element forward operator, analytic-gated to correlation 1.00 on a concentric-sphere problem, honestly does *not* beat the calibrated single-layer on the real coarse electrode geometry (dog: 0.54 vs 0.63 relative error); the null is reported. *Case 2 (4D-flow aortic pressure, incompressible Navier–Stokes).* A divergence-free velocity network denoises a real thoracic-aorta 4D-flow scan (raw divergence reduced $2.3\times$) and the relative pressure is forced out by the pressure-Poisson equation with the source and Neumann flux computed from the network's analytic derivatives rather than by finite differences at the lumen edge, which is what takes the recovered pressure from a non-physiological thousands of mmHg to a physiological 0.79 mmHg range for this unobstructed aorta, the same order as the clinical simplified-Bernoulli estimate (2.51 mmHg). The solve is analytic-gated to correlation 1.00 on a converging duct; the momentum-residual PINN that leaves pressure gauge-free is kept as the documented failed baseline. The common thread is methodological: separate the well-posed part (velocity, potentials) from the ill-posed part (pressure, inverse source), validate against a real gold standard or an exact gate, and publish the nulls.

***Index Terms*—physics-informed neural networks, electrocardiographic imaging, inverse problems, 4D-flow MRI, pressure-Poisson equation, Navier–Stokes, volume conduction, boundary element method, deep ensembles, uncertainty quantification, cardiac reconstruction.**

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Preprint, 2026. Code, derived reconstruction artifacts, and the interactive viewer are released under MIT at https://github.com/fsantibanezleal/CAOS_RES_CardioPINN; live at https://cardiopinn.fasl-work.com. This is a validated methodological result on real experimental data, not a clinically deployed system; raw datasets are used under their data-use agreements and are not redistributed.

I. Introduction

PHYSICS-INFORMED neural networks (PINNs) embed a governing differential equation in the training loss, so a network can fit sparse or noisy measurements while respecting the physics that relates them to unobserved fields [1]. Their most compelling use is not re-solving a forward problem a classical solver already handles, but recovering a field that *cannot* be measured from one that can [2]. Two cardiac problems have exactly this shape. In electrocardiographic imaging (ECGi), the clinician needs the heart-surface potential map to localize an arrhythmia, but a patient offers only the body-surface recording; the true heart-surface map, from an electrode cage, is a gold standard one never has in the clinic [5]. In vascular flow, the clinician needs the pressure drop across a stenosis, but 4D-flow MRI measures only velocity; pressure is obtained invasively by catheter [4]. In both, a governing equation ties the wanted field to the measured one.

This paper is deliberately real-data-only. A recurring failure mode in physics-informed reconstruction is to validate on synthetic data, where a network is scored against a field a solver produced under the same assumptions the network uses; such a result demonstrates code correctness, not a clinical capability. We therefore fit only real measured signals and validate only against a real measured gold standard or, where no gold standard can exist (pressure is never measured non-invasively), against an exact analytic gate that must pass before any real datum is trusted. We report two cases in two different physics domains and, crucially, the honest negative results each produced: a boundary-element forward operator that does not beat a simpler one on real geometry, and a momentum-residual PINN that cannot recover pressure at aortic Reynolds number. Both nulls are informative and are kept on the record.

II. Case 1: Electrocardiographic imaging (volume conduction)

A. Problem and forward operator

In the quasi-static approximation the torso is a volume conductor and the potential ϕ obeys Laplace's equation $\nabla \cdot (\sigma \nabla \phi) = 0$ between the heart surface and the body surface.

TABLE I
REAL ECGI VALIDATION AGAINST THE MEASURED CAGE POTENTIALS (IDENTICAL PIPELINE, NO PER-HEART RETUNING). RE: RELATIVE ERROR; CC: SPATIAL CORRELATION; UQ: FRACTION OF NODES WITHIN TWO STANDARD DEVIATIONS.

Dataset	Beat	RE	CC	UQ (2σ)
Human torso tank	Sinus	0.65	0.72	0.90
Human torso tank	Paced (PVP)	0.58	0.80	0.89
Human torso tank	AV-paced (AVP)	0.54	0.85	0.90
In-situ dog	Sinus	0.54	0.78	0.90

Discretized on the real geometry, the body-surface potentials b are a linear map of the heart-surface potentials x,

$$b = A\,x + \varepsilon, \tag{1}$$

where A is the transfer (lead-field) matrix and ε is measurement noise. Recovering x from b is severely ill-posed: the singular values of A decay rapidly, so noise in b is amplified without bound. We build A two ways: a single-layer point-source Green's-function approximation, and a full boundary-element operator (BEM) whose double-layer entries use the exact solid angle of a plane triangle [8].

B. Regularized inversion with a spatial prior and node uncertainty

The reconstruction is a Tikhonov problem with a graph-Laplacian prior L on the heart surface,

$$\hat{x} = \arg\min_{x} \, \|A\,x - b\|_2^2 + \lambda\,\|L\,x\|_2^2, \tag{2}$$

with λ chosen on the L-curve [9]. Because a single regularized point estimate carries no honest error bar, we draw an ensemble over realistic measurement-noise perturbations of b and re-solve, yielding a per-node predictive distribution [10]; the ensemble spread is recalibrated so that its two-standard-deviation band covers the true per-node error at the nominal rate.

C. Real validation on EDGAR

The pipeline is applied, with no per-heart retuning, to independent real EDGAR experiments [6]: a human torso tank (192 body electrodes $\to$ 256-node cage; sinus and two paced beats) and an in-situ dog (140 body $\to$ 1321-node epicardium; sinus). The recovered heart-surface potentials are compared to the real measured cage potentials with the standard relative error (RE) and spatial correlation (CC) [7] (Table I). The numbers are literature-consistent for torso-tank ECGi, the node uncertainty covers $\approx 90\%$ of nodes at two standard deviations, and the paced beats reconstruct better than sinus (higher CC), which is physically expected: a focal paced activation is easier to localize than a diffuse sinus wavefront.

D. Honest null: BEM does not beat the single-layer here

A more physically complete forward operator is not automatically a better one. The BEM is analytic-gated on a concentric-sphere problem (correlation 1.00, error halving per mesh refinement), so its implementation is correct. Yet on the real electrode geometry it does not beat the calibrated single-layer: it requires closed 2-manifold surfaces (the human torso-tank surface is open, so the BEM applies only to the dog case), and where it applies the coarse 140-node torso makes the reconstruction regularization-dominated (dog: single-layer RE 0.54 vs. BEM RE 0.63). Forward-operator fidelity is not the bottleneck at this electrode density; the single-layer stays the default, and the BEM is expected to matter only as electrode density and mesh closure improve. This null is reported, not hidden.

III. CASE 2: 4D-FLOW AORTIC PRESSURE (NAVIER–STOKES)

A. Problem and the pressure-Poisson route

For an incompressible Newtonian fluid the momentum equation is

$$\rho\Big(\partial_t v + (v\cdot\nabla)v\Big) = -\nabla p + \mu\nabla^2 v, \qquad \nabla\cdot v = 0. \tag{3}$$

Taking the divergence and using incompressibility eliminates the velocity time derivative from the elliptic part and yields a Poisson equation for pressure,

$$\nabla^2 p = S(v) = -\rho\,\nabla\cdot\Big((v\cdot\nabla)v\Big) + \rho\,\nabla\cdot(\partial_t v), \tag{4}$$

whose source is built from spatial derivatives of the measured velocity. Because that source is a product of derivatives, measurement noise (which violates incompressibility) is amplified. A physics-informed velocity step is therefore required first: a network fits the measured velocity while enforcing $\nabla\cdot v = 0$,

$$\mathcal{L} = \|v_\theta - v_{\text{meas}}\|_2^2 + \beta\,\|\nabla\cdot v_\theta\|_2^2, \tag{5}$$

producing a smooth divergence-free field whose analytic derivatives are clean. The pressure-Poisson source and the Neumann wall flux in (4) are computed from those analytic derivatives, *not* by finite differences at the lumen edge, where finite differences manufacture the worst artifacts. This analytic treatment is what takes the recovered pressure from a non-physiological thousands of mmHg to a physiological range [3]. The unsteady term $\partial_t v$ is differentiated exactly by a space-time network trained over the whole cardiac cycle.

B. The momentum-residual PINN failure (documented baseline)

A single network $(x, y, z, t) \to (u, v, w, p)$ trained on the raw momentum residual (the hidden-fluid-mechanics formulation [2]) does not recover pressure at aortic Reynolds number: pressure is gauge-free and only weakly coupled to the loss, so it stays near its initialization (on an analytic Poiseuille flow it recovered under 10% of the true gradient). The shipped method instead separates the well-posed part (velocity, strongly data-constrained) from the ill-posed part (pressure, solved by the elliptic equation). The failed approach is kept in the repository as the documented baseline.

TABLE II
REAL 4D-FLOW PRESSURE RECONSTRUCTION (THORACIC AORTA, PHILIPS, VENC 120 CM/S, 16 FRAMES).

Quantity	Value
Lumen voxels resolved	47,902
Peak velocity (denoised field)	0.791 m/s
Raw measured speed (noise-inflated)	≈ 1.6 m/s
Divergence reduction (denoiser)	$2.3\times$ ($25.4 \to 11.2$ s^{-1})
PPE relative-pressure range	0.79 mmHg
Clinical Bernoulli $4V_{\max}^2$	2.51 mmHg
Analytic gate, steady (converging duct)	corr 1.00; 4.74 vs. 4.73 mmHg
Analytic gate, unsteady (Poiseuille dv/dt)	corr 0.995
Phase-wrap aliasing corrected	27,863 samples
Noise-perturbation sensitivity	< 0.01 mmHg

C. Real validation with no invasive gold standard

No non-invasive pressure gold standard exists; the validation is therefore threefold (Table II). *An exact analytic gate*: on a converging-duct flow whose pressure drop is known analytically, the solve recovers 4.74 vs. 4.73 mmHg (correlation 1.00), and it runs in continuous integration before any real datum is trusted. *A physiological range on the real scan*: the recovered relative pressure spans 0.79 mmHg across the segment, small and physiological for an unobstructed aorta (the earlier three-frame finite-difference unsteady term had inflated this to ≈ 15 mmHg; the analytic space-time term, gated to dv/dt correlation 0.995, removes the inflation). *A clinical bracket*: the simplified-Bernoulli estimate $4\,V_{\max}^2 =$ 2.51 mmHg from the denoised peak velocity 0.791 m/s is the same order, exactly what an unobstructed aorta should show, while the field additionally reveals *where* the pressure varies, which a single Bernoulli number cannot.

D. An honest note on uncertainty

A deep ensemble that perturbs the measured velocity with realistic phase-contrast noise (5% of the venc) and re-runs the whole pipeline moves the recovered pressure by under 0.01 mmHg: the divergence-free denoiser makes the pressure essentially insensitive to velocity measurement noise. This is a genuine robustness, but it also means an ensemble over that noise gives a near-zero, uninformative uncertainty. The dominant uncertainty is instead the absent invasive gold standard, the lumen segmentation, and the unsteady-term approximation, none of which such an ensemble can quantify. We therefore report the robustness as a scalar and deliberately do not show a per-voxel pressure uncertainty map, which would be a misleadingly uniform near-zero field. (This is the opposite of Case 1, where the ill-posed inverse makes the ensemble uncertainty large and informative, so it is shown per node.)

IV. DISCUSSION

The two cases share one methodological spine. Each separates a well-posed part that data constrains (the body-surface potentials; the velocity field) from an ill-posed part that the physics forces out (the heart-surface potentials via a regularized inverse; the pressure via an elliptic equation). Each validates against something real: a measured gold standard where one exists (the ECGi cage), and an exact analytic gate where one cannot (the converging-duct pressure drop). And each reports what did not work: the BEM that does not beat the single-layer on real coarse geometry, and the momentum-residual PINN that leaves pressure gauge-free. The uncertainty treatment is case-appropriate rather than uniform: informative and shown per node where the inverse is ill-posed (ECGi), and honestly suppressed to a scalar where the denoiser makes it near-zero (4D-flow). Reporting the nulls and matching the uncertainty to the problem is what separates a validated methodological result from a demonstration.

V. LIMITATIONS AND CONCLUSION

These are validated methodological results on real experimental data, not clinically deployed tools. ECGi is shown on a torso tank and one in-situ animal, not a patient cohort; the 4D-flow pressure has no invasive truth to check its absolute magnitude against, so only its analytic gate, physiological range, divergence-free denoising, and Bernoulli bracket are claimed. Raw datasets are used under data-use agreements and are not redistributed; only the derived reconstruction artifacts are committed, and every number here is reproducible from them. The contribution is a demonstration that, with a real-data-only discipline and honest nulls, physics-informed reconstruction can recover two clinically meaningful, unmeasurable cardiac fields, ECGi heart-surface potentials and 4D-flow aortic pressure, from the non-invasive measurements that are tied to them by a governing equation.

DATA AND CODE AVAILABILITY

Code, the derived reconstruction artifacts (the ECGi catalogue and the 4D-flow pressure trace), and the interactive viewer are at https://github.com/fsantibanezleal/CAOS_RES_CardioPINN under the MIT license; a static viewer is live at https://cardiopinn.fasl-work.com. ECGi data are from the EDGAR repository (Consortium for ECG Imaging); the 4D-flow scan is used under its data-use agreement. Raw datasets are not redistributed.

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